**NPTEL Online Certification Course**
**Fundamentals of Artificial Intelligence**
**MOOCs; July - Dec 2026**
**Indian Institute of Technology Guwahati**
**Assignment No. 7: Detailed Solution**

10 Marks

Each question carries 01 Mark each.

Q1. A logical agent differs from a probabilistic agent because a probabilistic agent _____.

A. uses binary truth values.
B. uses graded belief values.
C. uses deterministic rules only.
D. always produces exact answers.

**Correct Answer: (B)**

Q2. Within Bayesian Decision Making, probability provides a formal way of representing _____ in the presence of incomplete knowledge.

A. information
B. randomness
C. belief
D. uncertainty.

**Correct Answer: (D)**

Q3. Bayesian Networks assume that all direct probabilistic dependencies are explicitly represented by the network arcs. The assumption that each node is conditionally independent of its non-descendants given its parents is known as the _____.

A. Bayesian Dependency
B. Local Markov Property
C. Bayesian Property
D. Arc Consistency

**Correct Answer: (B)**

Q4. In a Bayesian Network, if nodes X and Y are d-separated by Z, then X and Y are _____ given Z.

A. conditionally dependent
B. conditionally independent
C. marginally independent
D. dependent

**Correct Answer: (B)**

Q5. A fair die is rolled once. Let event A = obtaining an even number.
What is P(A)?

Indian Institute of Technology Guwahati

A. 1/3
B. 1/2
C. 2/3
D. 1/6

**Correct Answer: (B)**

**Detailed Solution:**

For one dice, n(S) = 6.

Events favourable for even numbers n(A) = 3

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{6} = \mathbf{\frac{1}{2}}$$

Q6. A disease affects 5% of a population.
A diagnostic test correctly identifies an infected person with probability 0.90.
What is the probability that a randomly selected individual is both infected and correctly identified?

A. 0.045
B. 0.050
C. 0.090
D. 0.950

**Correct Answer: (A)**

**Detailed Solution:**

Probability of an individual affected by the disease, P(A) = 0.05

Probability of correct identification of the disease using diagnostic test, P(B) = 0.90

So, Probability that a randomly selected individual is both infected and correctly identified,

$$P(A \cap B) = P(A).P(B) = (0.05) * (0.90) = \mathbf{0.045}$$

Q7. In a class, 60% of students pass Mathematics.
Among those who pass Mathematics, 80% pass AI.
What is the probability that a randomly selected student passes both Mathematics and AI?

A. 0.48
B. 0.60
C. 0.80
D. 0.32

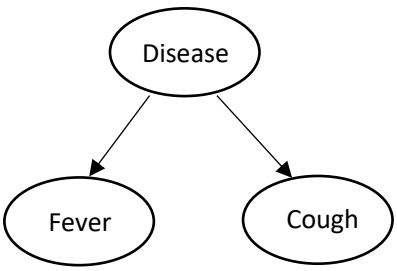
**Correct Answer: (A)**

**Detailed Solution:**

Probability that a randomly selected student passes both Mathematics and AI is given by

$$P(A).P(B) = 0.6 * 0.8 = \mathbf{0.48}$$

Q8. Consider the Bayesian Network shown.
Given: P(Disease = T) = 0.20
P(Fever = T| Disease = T) = 0.80
P(Fever = T| Disease = F) = 0.10




Indian Institute of Technology Guwahati

What is the probability that a patient has fever?

A. 0.18
B. 0.30
C. 0.24
D. 0.50
**Correct Answer: (C)**
**Detailed Solution:**

$$P(Fever = T) = P(Disease = T) * P(Fever = T|Disease = T) + P(Disease = F) * P(Fever = T|Disease = F)$$

$$P(Fever = T) = 0.20 * 0.80 + (1 - 0.20) * 0.10 = \mathbf{0.24}$$

Q9. Machine A produces 60% of items with defect rate 2%.
Machine B produces 40% of items with defect rate 5%.
An item selected at random is found defective.
What is the probability that it was produced by Machine B?

A. 0.200
B. 0.625
C. 0.375
D. 0.500
**Correct Answer: (B)**
**Detailed Solution:**
Let's consider the following,
A: Item is produced by machine A
B: Item is produced by machine B
D: Item is defective
So, P(A) = 0.6; P(B) = 0.4; P(D|A) = 0.02; P(D|B) = 0.05.
Using Bayes' Theorem,

$$P(B|D) = \frac{P(D|B) * P(B)}{P(D|A) * P(A) + P(D|B) * P(B)}$$

$$P(B|D) = \frac{(0.05 * 0.40)}{(0.02 * 0.60) + (0.05 * 0.40)} = \mathbf{0.625}$$

Q10. A decision-maker has two possible actions.

| Action | Utility if Success | Utility if Failure |
|---|---|---|
| A1 | 100 | -20 |
| A2 | 80 | -5 |

Probability of success = 0.6
Probability of failure = 0.4

Which action should a rational agent choose?

Indian Institute of Technology Guwahati

A. A1
B. A2
C. Both equal
D. Cannot be determined

**Correct Answer: (A)**

**Detailed Solution:**

The expected utility (EU) is given by,

$$EU(A_1) = P(S) * U(A_1, S) + P(F) * U(A_1, F)$$
$$EU(A_1) = (0.6 * 100) + [0.4 * (-20)] = 52$$

$$EU(A_2) = P(S) * U(A_2, S) + P(F) * U(A_2, F)$$
$$EU(A_2) = 0.6 * 80 + 0.4 * (-5) = 46$$

Since, EU($A_1$) > EU($A_2$). So, a rational agent should choose action $A_1$.

Indian Institute of Technology Guwahati